



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**NATIONAL
SENIOR CERTIFICATE
NASIONALE
SENIOR SERTIFIKAAT**

GRADE 12/GRAAD 12

TECHNICAL MATHEMATICS P1/TEGNIESE WISKUNDE V1

NOVEMBER 2022

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

MARKING CODES/NASIENKODES	
A	Accuracy/Akkuraatheid
CA	Consistent accuracy/Volgehoue akkuraatheid
M	Method/Metode
R	Rounding/Afronding
NPR	No penalty for rounding/Geen penalisering vir afronding nie
NPU	No penalty for units omitted / Geen penalisering vir eenhede weggelaat nie
S	Simplification/Vereenvoudiging
SF	Substitution in correct formula/Vervanging in korrekte formule
PR	Penalty for rounding/Penalisasie vir afronding

**These marking guidelines consist of 19 pages.
Hierdie nasienriglyne bestaan uit 19 bladsye.**

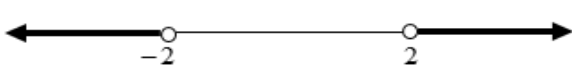
NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- Consistent accuracy marking to be applied where indicated.
- # Shows questions where Tolerance Range will be applied:
Q 3.3 ; Q 5.3 ; Q 6.2.3 ; Q 9.1.3

LET WEL:

- Indien 'n kandidaat 'n vraag TWEE keer beantwoord, sien slegs die EERSTE poging na.
- Volgehoue akkuraatheidnasien moet toegepas word soos aangedui.
- # Toon vrae waar Toleransie Wydte (Verdraagsaamheids omvang) toegepas word:
V 3.3 ; V 5.3 ; V 6.2.3 ; V 9.1.3

QUESTION/VRAAG 1

1.1.1	$x(7 + x) = 0$ $x = 0$ or/of $x = -7$	$\checkmark x = 0$ $\checkmark x = -7$	A A (2)
1.1.2	$4x^2 - 5x - 4 = 0$ $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $= \frac{-(-5) \pm \sqrt{(-5)^2 - 4(4)(-4)}}{2(4)}$ OR/OF $= \frac{-(5) \pm \sqrt{(5)^2 - 4(-4)(4)}}{2(-4)}$ $\therefore x \approx 1,80$ or / of $x \approx -0,55$	$\checkmark$ SF $\checkmark$ positive x- value / positiewe x- waarde $\checkmark$ negative x- value / negatiewe x- waarde NPR AO full marks /volpunte	A CA CA (3)
1.1.3	$2x^2 - 8 > 0$ $2(x^2 - 4) > 0$ $2(x - 2)(x + 2) > 0$ OR/OF $(x - 2)(x + 2) > 0$ Critical values/kritiese waardes: 2 and/en -2 $\therefore x < -2$ or/of $x > 2$ OR/OF $x \in (-\infty; -2)$ or/of $x \in (2; \infty)$ OR/OF 	$\checkmark$ factors/faktore Formula / Formule $\checkmark$ Both critical values/ beide kritiese waardes $\checkmark$ notation/notasie	A CA A AO full marks /volpunte (3)

1.2	$y = 5x - 2$ and/en $y = x^2 + 4x - 8$ $x^2 + 4x - 8 = 5x - 2$ $x^2 - x - 6 = 0$ $(x - 3)(x + 2) = 0$ OR/OF $x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-6)}}{2(1)}$ $\therefore x = 3$ or/of $x = -2$ OR/OF $y = 5(3) - 2$ or/of OR / OF $y = (3)^2 + 4(3) - 8$ or/of $y = 5(-2) - 2$ $y = (-2)^2 + 4(-2) - 8$ $\therefore y = 13$ or/of $y = -12$ OR/OF $x = \frac{y + 2}{5}$ $y = \left(\frac{y + 2}{5}\right)^2 + 4\left(\frac{y + 2}{5}\right) - 8$ $y = \frac{y^2 + 4y + 4}{25} + \frac{4y + 8}{5} - 8$ $25y = y^2 + 4y + 4 + 20y + 40 - 200$ $y^2 - y - 156 = 0$ $(y - 13)(y + 12) = 0$ OR/OF $y = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-156)}}{2(1)}$ $\therefore y = 13$ or / of $y = -12$ $x = \frac{13 + 2}{5}$ or / of $x = \frac{-12 + 2}{5}$ $\therefore x = 3$ and/en $x = -2$	✓ equating/gelykstel A ✓ standard form/ standaardvorm CA ✓ factors or formula/ faktore of formule CA ✓ both x-values/ beide x- waardes CA ✓ both y-values/ beide y-waardes CA OR/OF ✓ substitution/ vervanging A ✓ standard vorm/ standaardvorm CA ✓ factors or formula/ faktore of formule CA ✓ both y-value/ beide y-waardes CA ✓ both x values/ beide x-waardes CA (5)
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1.3.1	$SV = \frac{\pi d^2 \times L}{4}$ $4SV = \pi d^2 \times L$ $\therefore L = \frac{4SV}{\pi d^2} \quad \text{OR/OF} \quad \therefore L = \frac{SV}{\frac{\pi d^2}{4}}$	✓ M A ✓ <i>L subject/L-onderwerp</i> CA AO full marks /volpunte (2)
1.3.2	$L = \frac{4SV}{\pi d^2}$ $= \frac{4 \times 1020,5}{\pi \times (10)^2} \quad \text{OR / OF} \quad = \frac{4 \times 1020,5}{3,14 \times (10)^2}$ $\approx 13 \text{ cm}$ OR/OF $L = \frac{SV}{\frac{\pi d^2}{4}}$ $= \frac{1020,5}{\frac{\pi (10)^2}{4}}$ $\approx 13 \text{ cm}$ OR/OF $SV = \frac{\pi d^2 \times L}{4}$ $1020,5 = \frac{\pi (10)^2 \times L}{4}$ $L \approx 13 \text{ cm}$	✓ SF CA ✓ value of <i>L/</i> <i>waarde v L</i> CA OR/OF ✓ SF CA ✓ value of <i>L/</i> <i>waarde v L</i> CA OR/OF ✓ SF A ✓ value of <i>L/</i> <i>waarde v L</i> CA PR in this question only/ slegs in hierdie vraag (2)

1.4.1	<table><tr><td></td><td>2⁵</td><td>2⁴</td><td>2³</td><td>2²</td><td>2¹</td><td>2⁰</td><td></td></tr><tr><td>P</td><td></td><td></td><td>1</td><td>0</td><td>1</td><td>0</td><td>=10</td></tr></table>		2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰		P			1	0	1	0	=10	✓ 10	A (1)																
	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰																													
P			1	0	1	0	=10																												
1.4.2	1010₂ × 10000₂ = 10100000₂ OR/OF <table><tr><td></td><td>2⁵</td><td>2⁴</td><td>2³</td><td>2²</td><td>2¹</td><td>2⁰</td><td></td></tr><tr><td>Q</td><td></td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>=16</td></tr></table> P × Q = 10 × 16 = 160 = 10100000₂ <table><tr><td>2⁷</td><td>2⁶</td><td>2⁵</td><td>2⁴</td><td>2³</td><td>2²</td><td>2¹</td><td>2⁰</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>		2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰		Q		1	0	0	0	0	=16	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	1	0	1	0	0	0	0	0	✓✓ AO full marks /volpunte ✓ 16 ✓ 10100000 ₂	A A CA (2)
	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰																													
Q		1	0	0	0	0	=16																												
2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰																												
1	0	1	0	0	0	0	0																												
			[20]																																

QUESTION/VRAAG 2

2.1.1	$x^2 - 2x + 6 = 0$ $\Delta = b^2 - 4ac$ $= (-2)^2 - 4(1)(6)$ $= -20$	✓ SF ✓ S AO full marks /volpunte (2)
2.1.2	Non-real/Imaginary/Nie-reël/imaginêr	✓ non-real/imaginary/ <i>nie-reël/imaginêr</i> CA (1)
2.2	$x^2 + 2x + k = 0$ $\Delta = (2)^2 - 4(1)(k)$ $(2)^2 - 4(1)(k) \geq 0$ $4 - 4k \geq 0$ $k \leq 1$	✓ SF A ✓ $\Delta \geq 0$ A ✓ value(s) of k / <i>waarde(s) van k</i> CA (3)
		[6]

QUESTION/VRAAG 3

3.1.1	$\frac{8x^3y^2}{16xy^4}$ $= \frac{x^2}{2y^2}$	$\checkmark x^2$ or/of $\frac{1}{2}x^2$ A $\checkmark 2y^2$ or/of y^2 A (2)
3.1.2	$\frac{\sqrt{48} + \sqrt{12}}{\sqrt{27}}$ $= \frac{4\sqrt{3} + 2\sqrt{3}}{3\sqrt{3}} \text{ OR/OF } = \frac{2^2\sqrt{3} + 2\sqrt{3}}{3\sqrt{3}} \text{ OR/OF}$ $= \frac{6\sqrt{3}}{3\sqrt{3}} = \frac{\sqrt{3}(4+2)}{3\sqrt{3}} = \frac{3^{\frac{1}{2}}(4+2)}{3 \cdot 3^{\frac{1}{2}}}$ $= 2$ OR/OF $\frac{\sqrt{48} + \sqrt{12}}{\sqrt{27}}$ $= \frac{4\sqrt{3}}{3\sqrt{3}} + \frac{2\sqrt{3}}{3\sqrt{3}}$ $= \frac{6}{3}$ $= 2$	$\checkmark$ simplified surd forms/ vereenvoudigde wortelvorme A $\checkmark$ S CA $\checkmark$ S CA OR/OF $\checkmark$ simplified surd forms/ vereenvoudigde wortelvorme A $\checkmark$ S CA $\checkmark$ S CA AO one mark/slegs een punt (3)
3.2.1	$\log 25$ $= \log 5^2 \text{ OR/OF } 2\log 5 \text{ OR/OF } \log 5 + \log 5$ $= 2m$	$\checkmark$ log(exp) property/ log(eksp.)eienskap A $\checkmark$ S CA AO full marks /volpunte (2)
3.2.2	$\log 2$ $= \log\left(\frac{10}{5}\right) \text{ OR/OF } = \log 2 + \log 5 - \log 5$ $= \log 10 - \log 5$ $= 1 - m$	$\checkmark$ M A $\checkmark$ log property/ eienskap A $\checkmark$ S CA AO full marks /volpunte (3)

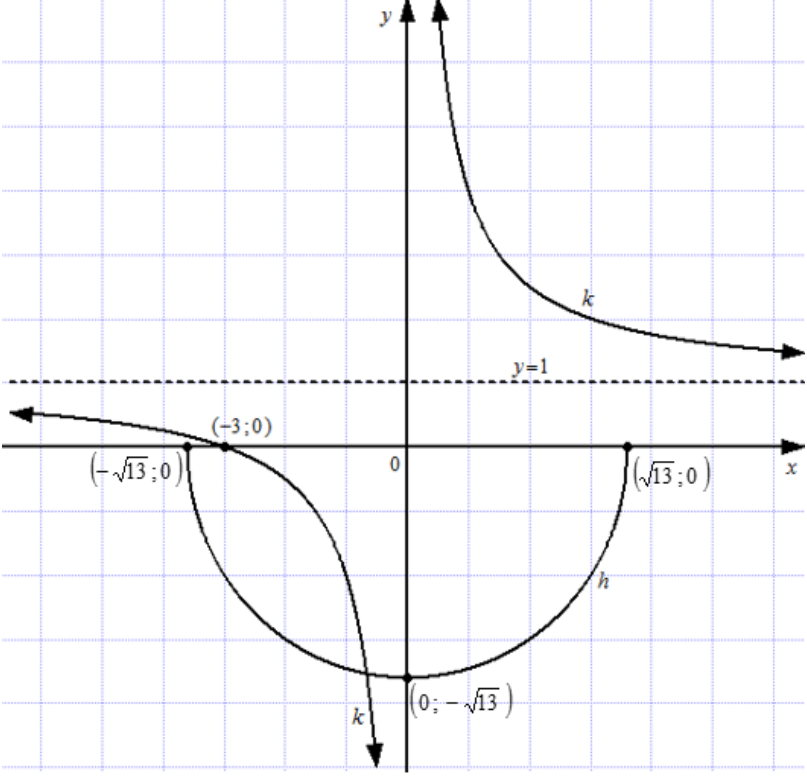
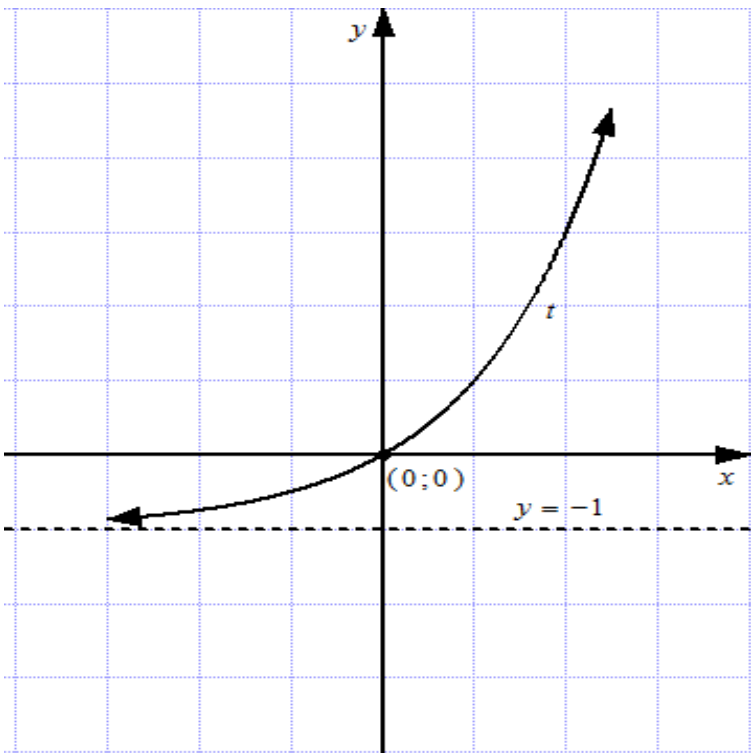
3.3	$\log_2(x+3) - 3 = -\log_2(x-4)$ $\log_2(x+3) + \log_2(x-4) = 3$ $\log_2(x+3)(x-4) = 3$ $(x+3)(x-4) = 2^3 = 8$ $x^2 - x - 12 = 8$ $x^2 - x - 20 = 0$ $(x+4)(x-5) = 0 \quad \text{OR/OF}$ $x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-20)}}{2(1)}$ $\therefore x = 5 \quad \text{or/of} \quad x \neq -4$ OR/OF $\log_2(x+3) - 3 = -\log_2(x-4)$ $\log_2(x+3) - 3\log_2 2 = \log_2(x-4)^{-1}$ $\log_2(x+3) - \log_2 8 = \log_2(x-4)^{-1}$ $\log_2 \frac{(x+3)}{8} = \log_2 \frac{1}{(x-4)}$ $\frac{x+3}{8} = \frac{1}{x-4}$ $(x+3)(x-4) = 1 \times 8$ $x^2 - x - 12 = 8$ $x^2 - x - 20 = 0$ $(x+4)(x-5) = 0 \quad \text{OR/OF}$ $x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-20)}}{2(1)}$ $\therefore x = 5 \quad \text{or/of} \quad x \neq -4$ OR/OF	✓ log property/-eienskap A ✓ exponential form/ eksponentvorm CA ✓ standard form/ standaardvorm CA ✓ factors or formula/ faktore of formule CA ✓ correct value of/ korrek waarde van x CA OR/OF ✓ log property/-eienskap A ✓ log property/-eienskap CA ✓ standard form/ standaardvorm CA ✓ factors or formula/ faktore of formule CA ✓ correct value of/ korrek waarde van x CA OR/OF
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	$\log_2(x+3) = 3 - \log_2(x-4)$ $\log_2(x+3) = \log_2 8 - \log_2(x-4)$ $\log_2(x+3) = \log_2 \frac{8}{(x-4)}$ $\frac{x+3}{1} = \frac{8}{x-4}$ $(x+3)(x-4) = 8$ $x^2 - x - 12 = 8$ $x^2 - x - 20 = 0$ $(x+4)(x-5) = 0 \quad \text{OR/OF}$ $x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-20)}}{2(1)}$ $\therefore x = 5 \text{ or/of } x \neq -4$	✓ log property/-eienskap A ✓ log property/-eienskap CA ✓ standard form/ standaardvorm CA ✓ factors or formula/ faktore of formule CA ✓ correct value of/ korrek waarde van x CA (5)
3.4.1	$\bar{z} = -1 - 3i$	✓ conjugate/toegevoegde A (1)
3.4.2	$z_2 = \sqrt{2} \operatorname{cis} 135^\circ = \sqrt{2} \cos 135^\circ + i\sqrt{2} \sin 135^\circ$ $= -1 + i$	✓ expansion/uitbreiding A ✓ $-1 + i$ A AO full marks /volpunte (2)
3.4.3	$z_1 - z_2$ $= -1 + 3i - (-1 + i) \text{ OR/OF } -1 + 3i - \sqrt{2} \operatorname{cis} 135^\circ$ $= 2i$	✓ substitution/vervanging CA ✓ S CA AO full marks /volpunte (2)
3.5	$x + yi - (1 - i) = 4 + 5i$ $x + yi - 1 + i = 4 + 5i$ $x + yi = 5 + 4i$ $\therefore x = 5 \text{ and/en } y = 4$ OR/OF $x + yi - (1 - i) = 4 + 5i$ $x + yi - 1 + i = 4 + 5i$ $(x-1) + (y+1)i = 4 + 5i$ $x - 1 = 4 \text{ and/en } y + 1 = 5$	✓ S A ✓ S CA ✓ value of x/waarde van x CA ✓ value of y/waarde van y CA OR/OF ✓ S A ✓ S CA ✓ value of x/waarde van x CA ✓ value of y/waarde van y CA (4)
		[24]

QUESTION/VRAAG 4

4.1.1(a)	$y \geq -9$ OR/OF $y \in [-9; \infty)$ OR/OF $-9 \leq y < \infty$	✓ $y \geq -9 / y \in [-9; \infty) /$ $-9 \leq y < \infty$	A (1)
4.1.1(b)	Q(4; -5)	✓ $x = 4$ ✓ $y = -5$	A A (2)
4.1.2(a)	$f(x) = x^2 - 4x - 5$ x – int s./ afsn. : $x^2 - 4x - 5 = 0$ $(x+1)(x-5) = 0$ OR/OF $x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(-5)}}{2(1)}$ $x = -1$ or / of $x = 5$	✓ $y = 0$ ✓ factors/form/ faktore/formule ✓ both values of x / beide waardes van x	A A CA AO full marks /volpunte (3)
4.1.2(b)	The length of/lengete van AB = 6 units/eenhede	✓ 6	CA (1)
4.1.3	$g(x) = mx + c$ through/deur A(-1; 0); Q(4; -5) $m = \frac{y_1 - y_2}{x_1 - x_2} = \frac{-5 - 0}{4 - (-1)}$ $= -1$ $g(x) = -1x + c$ OR/OF $y - y_1 = -1(x - x_1)$ subst./ verv. : (-1; 0) $0 = -1(-1) + c$ $y - 0 = -1(x + 1)$ $c = -1$ OR/OF subst./ verv. : (4; -5) $-5 = -1(4) + c$ OR/OF $y - (-5) = -1(x - 4)$ $c = -1$	✓ SF ✓ $m = -1$ Must be negative/ Moet negatief wees ✓ $c = -1$	CA CA CA (3)

4.1.4	$x < 5$; $x \neq -1$ OR/OF $x \in (-\infty; 5)$; $x \neq -1$ OR/OF $x \in (-\infty; -1)$ or / of $(-1; 5)$	✓ $x < 5$ ✓ $x \neq -1$ OR/OF ✓ $(-\infty; 5)$ ✓ $x \neq -1$ OR/OF ✓ $(-\infty; -1)$ ✓ $(-1; 5)$	CA CA CA CA CA CA (2)
4.2.1	$-\sqrt{13} \leq x \leq \sqrt{13}$ OR/OF $x \in [-\sqrt{13}; \sqrt{13}]$ OR/OF $-3,61 \leq x \leq 3,61$ OR/OF $x \in [-3,61; 3,61]$ OR/OF $x \geq -3,61$ and / en $x \leq 3,61$ OR/OF $x \geq -\sqrt{13}$ and / en $x \leq \sqrt{13}$	✓ critical values/ kritiese waardes ✓ notation/notasie	A A (2)
4.2.2(a)	$x = 0$; $y = 1$	✓ $x = 0$ ✓ $y = 1$	A A (2)
4.2.2(b)	$0 = \frac{3}{x} + 1$ $-1 = \frac{3}{x}$ $x = -3$	✓ $y = 0$ ✓ value of x /waarde van x	A A AO full marks /volpunte (2)

4.2.3		<i>k</i>: ✓ shape/vorm A ✓ x-int./afsn. CA ✓ asymptote/asimptote CA <i>h</i>: ✓ shape/vorm A ✓ both/beide x-int./afsn. CA (5)
4.3		✓ shape/vorm A ✓ (0;0) A ✓ asymptote/asimptoot A (3)
	[26]	

QUESTION/VRAAG 5

5.1	$A = P(1 + i)^n$ $= R\ 8\ 000(1 + 13\%)^3$ $\approx R\ 11\ 543,18$	✓ F ✓ SF ✓ S NPR	A A CA (3)
5.2.1	10 621	✓ 10 621	A (1)
5.2.2	g	✓ g	A (1)
5.2.3	$A = P(1 - i)^n$ $3459 = 10\ 621(1 - 0,128)^n$ $\frac{3459}{10\ 621} = (0,872)^n$ $n = \frac{\log\left(\frac{3459}{10\ 621}\right)}{\log(0,872)}$ OR/OF $n = \log_{0,872} \frac{3459}{10\ 621}$ $\therefore n \approx 8,1905$ years/jaar	✓ F ✓ SF ✓ S ✓ log form/-vorm ✓ value of/waarde van n NPR	A A CA CA CA (5)
5.3 #	Value of investment end of 24 months/ <i>Waarde van belegging einde van 24 maande:</i> $\approx R\ 22\ 543,19552$ Invested/ <i>Belê</i> R5 000 $R\ 22\ 543,19552 + R\ 5\ 000 \approx R\ 27\ 543,19552$ Value of investment end of 5 years/ <i>Waarde van belegging einde van 5 jaar:</i> $A = P(1 + i)^n = 27\ 543,19552 \left(1 + \frac{5\%}{2}\right)^{3 \times 2}$ $\approx R\ 31\ 941,66$ $\therefore R\ 31\ 941,66 < R\ 35\ 000$ He will NOT have enough money/ <i>Hy sal NIE</i> <i>genoeg geld hê NIE.</i> OR/OF	$A = P(1 + i)^n = R\ 20\ 000 \left(1 + \frac{6\%}{12}\right)^{2 \times 12}$ ✓ SF ✓ M + R5 000 ✓ SF ✓ R 31 941, 66 ✓ conclusion/gevolgtrekking OR/OF	A A CA CA CA

	$A = R20\,000 \left(1 + \frac{6\%}{12}\right)^{2 \times 12} \left(1 + \frac{5\%}{2}\right)^{2 \times 3}$ $+ R5\,000 \left(1 + \frac{5\%}{2}\right)^{2 \times 3}$ $\approx R31\,941,66$ $\therefore R31\,941,66 < R35\,000$ He will NOT have enough money/Hy sal NIE genoeg geld hê NIE. OR/OF $P = \frac{R35\,000}{\left(1 + \frac{6\%}{12}\right)^{2 \times 12} \left(1 + \frac{5\%}{2}\right)^{2 \times 3}}$ $= \frac{5\,000}{\left(1 + \frac{6\%}{12}\right)^{2 \times 12}}$ $\approx R22\,339,68$ $\therefore R20\,000 < R22\,339,68$ He will NOT have enough money/Hy sal NIE genoeg geld hê NIE.	$\checkmark \left(1 + \frac{6\%}{12}\right)^{2 \times 12} \quad \mathbf{A}$ $\checkmark \left(1 + \frac{5\%}{2}\right)^{2 \times 3} \quad \mathbf{A}$ $\checkmark \mathbf{M} + R5\,000 \quad \mathbf{A}$ $\checkmark R31\,941,66 \quad \mathbf{CA}$ $\checkmark \text{conclusion/gevolgtrekking} \quad \mathbf{CA}$ OR/OF $\checkmark \left(1 + \frac{6\%}{12}\right)^{2 \times 12} \quad \mathbf{A}$ $\checkmark \left(1 + \frac{5\%}{2}\right)^{2 \times 3} \quad \mathbf{A}$ $\checkmark \mathbf{M} - R5\,000 \quad \mathbf{A}$ $\checkmark R22\,339,68 \quad \mathbf{CA}$ $\checkmark \text{conclusion/gevolgtrekking} \quad \mathbf{CA}$ NPR (5)
		[15]

QUESTION/VRAAG 6

6.1	$f(x) = 5 - 8x$ $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ $= \lim_{h \rightarrow 0} \frac{5 - 8(x+h) - (5 - 8x)}{h}$ $= \lim_{h \rightarrow 0} \frac{5 - 8x - 8h - 5 + 8x}{h}$ $= \lim_{h \rightarrow 0} \frac{-8h}{h}$ $= \lim_{h \rightarrow 0} (-8)$ $f'(x) = -8$	✓ definition/definisie A ✓ SF A ✓ S CA ✓ S CA ✓ -8 CA (5)
6.2.1	$f(x) = 3x^5 + \pi x$ $f'(x) = 15x^4 + \pi$	✓ $15x^4$ A ✓ π A (2)
6.2.2	$y = x^2(4x - 2x^{-1})$ $y = 4x^3 - 2x$ $\frac{dy}{dx} = 12x^2 - 2$	✓ S A ✓ $12x^2$ CA ✓ -2 CA (3)
6.2.3 #	$D_x \left[\sqrt[5]{x^4} - \frac{2}{5x^2} + 8t^4x \right]$ $= D_x \left[x^{\frac{4}{5}} - \frac{2}{5}x^{-2} + 8t^4x \right]$ $= \frac{4}{5}x^{-\frac{1}{5}} + \frac{4}{5}x^{-3} + 8t^4$	✓ $x^{\frac{4}{5}}$ A ✓ $-\frac{2}{5}x^{-2}$ A ✓ $\frac{4}{5}x^{-\frac{1}{5}}$ CA ✓ $\frac{4}{5}x^{-3}$ CA ✓ $8t^4$ CA (5)
6.3.1	$g'(x) = 12x + 3$ $\therefore 12(p) + 3 = -21$ $12p = -24$ $p = -2$	✓ derivative of/afgeleide van g A ✓ equat.deriv/afgeleide = -21 A ✓ $p = -2$ CA (3)
6.3.2	$y = 6(-2)^2 + 3(-2)$ $\therefore y = 18$ $y = mx + c$ OR/OF $y - y_1 = m(x - x_1)$ $18 = -21(-2) + c$ $y - 18 = -21(x + 2)$ $c = -24$ $y - 18 = -21 - 42$ $\therefore y = -21x - 24$	✓ y coordinate/y-koördinaat CA ✓ SF CA ✓ eqn. of tangent/vergelyk. van raaklyn CA (3)
		[21]

7.5	$-3 \leq x \leq 1$ OR/OF $x \in [-3 ; 1]$ OR/OF $x \geq -3$ and/en $x \leq 1$	✓ critical values/ <i>kritiese waardes</i> ✓ notation/notasie CA A (2)
7.6	$(1; -2)$	✓ x coordinate/ -koördinaat ✓ y coordinate/ -koördinaat A A (2)
		[15]

QUESTION/VRAAG 8

8.1	37,5 °C	✓ 37,5 °C NPU	A (1)
8.2	$T'(t) = 7 - t$ $T'(4) = 7 - (4)$ $= 3 \text{ °C/s}$	✓ derivative/ <i>afgeleide</i> ✓ Subst./verv. $t = 4$ ✓ 3 °C/s NPU	A A CA (3)
8.3	$7 - t = 0$ OR/OF $t = \frac{-(7)}{2\left(-\frac{1}{2}\right)}$ $\therefore t = 7s$ $T(7) = 37,5 + 7(7) - 0,5(7)^2 = 62 \text{ °C}$	✓ $7 - t = 0 / t = \frac{-(7)}{2\left(-\frac{1}{2}\right)}$ ✓ $t = 7s$ ✓ 62 °C NPU	A CA CA (3)
8.4	$7 - t < 0$ $\therefore t > 7$ $7 < t \leq 10$ OR/OF $t \in (7;10]$ OR/OF $t > 7$ and/en $t \leq 10$	✓ $t > 7$ ✓ restricting to 10/ <i>beperking tot 10</i> AO full marks /volpunte	CA A (2)
			[9]

QUESTION/VRAAG 9

9.1.1	$\int 3x^{-1} dx$ $= 3 \ln x + C$	✓ 3 ln x A ✓ C A (2)
9.1.2	$\int (4 + 2^{-x}) dx$ $= 4x - \frac{2^{-x}}{\ln 2} + C$ OR/OF $4x + \frac{2^{-x}}{\ln 2^{-1}} + C$ OR/OF $= 4x - \frac{\left(\frac{1}{2}\right)^x}{\ln 2} + C$ OR/OF $4x + \frac{\left(\frac{1}{2}\right)^x}{\ln \frac{1}{2}} + C$	✓ 4x A ✓ value of 2nd integral/ waarde van die tweede integraal A If C is omitted : NO penalty/ Indien C weggelaat is : GEEN penalisering (2)
9.1.3 #	$\int \frac{8x^4 - x^2}{2x} dx$ $= \int \left(4x^3 - \frac{1}{2}x \right) dx$ $= x^4 - \frac{1}{4}x^2 + C$	✓ S A ✓ x^4 CA ✓ $-\frac{1}{4}x^2$ CA No simplification: 0 marks/ Geen vereenvoudiging: 0 punte No penalty if C omitted/Geen penalisering indien C uitgelaat (3)
9.2	$A = \int_2^4 h(x) dx = \int_2^4 (-x^2 + 2x + 8) dx$ $= \left[-\frac{x^3}{3} + x^2 + 8x \right]_2^4$ $= \left[-\frac{(4)^3}{3} + (4)^2 + 8(4) \right] - \left[-\frac{(2)^3}{3} + (2)^2 + 8(2) \right]$ $= \frac{28}{3} \approx 9,33$ $\therefore \frac{\left(\frac{28}{3}\right)}{36} \times 100\% \approx 25,93\%$ OR/OF $20\% \times 36 = \left(\frac{36}{5}\right) = 7,2$ $25,93\% > 20\%$ $7,2 < 9,33$ $\therefore$ The learner's statement is NOT correct/ Die leerder se bewering is NIE korrek nie.	✓ Area notation using integrals/Oppervlakte- notasie met gebruik van integrale A ✓ $-\frac{x^3}{3} + x^2 + 8x$ A ✓ ✓SF CA ✓ area /oppervlakte CA ✓ % calculation / berekening CA ✓ conclusion/ Afleiding. CA

	OR/OF	OR/OF
	$A = 36 - \int_{-2}^2 h(x) dx = 36 - \int_{-2}^2 (-x^2 + 2x + 8) dx$ $= 36 - \left[-\frac{x^3}{3} + x^2 + 8x \right]_{-2}^2$ $= 36 - \left[-\frac{(2)^3}{3} + (2)^2 + 8(2) \right] - \left[-\frac{(-2)^3}{3} + (-2)^2 + 8(-2) \right]$ $= \frac{28}{3} \approx 9,33$ $\therefore \left(\frac{28}{3} \right) \times 100\% \approx 25,93\% \text{ OR/OF } 20\% \times 36 = \left(\frac{36}{5} \right) = 7,2$ $25,93\% > 20\% \qquad 7,2 < 9,33$ $\therefore$ The learner's statement is NOT correct/ Die leerder se bewering is NIE korrek nie.	✓ Area notation using integrals/ <i>Oppervlakte - notasie met gebruik van integrale</i> A ✓ $-\frac{x^3}{3} + x^2 + 8x$ A ✓ ✓ SF CA ✓ area/oppervlakte CA ✓ % calculation / <i>berekening</i> CA ✓ conclusion/ <i>afleiding</i> CA (7)
		[14]
TOTAL/TOTAAL: 150		